

Visualizing Spatial Distribution of Poly(*n*-Alkyl Acrylates) Aggregates in Pressure Sensitive Adhesive Films

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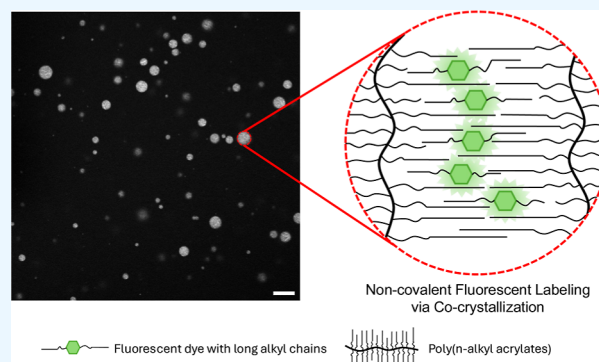
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ABSTRACT: Poly(*n*-alkyl acrylates) exhibit tunable melting temperatures, which can be precisely adjusted by varying the length and composition of their alkyl side chains. These polymers serve as promising thermoresponsive additives in pressure-sensitive adhesive (PSA) systems, particularly for medical tapes designed for individuals with sensitive skin. However, the mechanism behind their heat-triggered debonding remains unclear due to the chemical similarity between the additive and the hybrid rubber-acrylic PSA matrix, which hinders contrast in conventional analytical techniques. In this study, we present a noncovalent fluorescent labeling strategy that leverages the cocrystallization of long-alkyl-chain dyes with semicrystalline additives to selectively visualize additive distribution within the matrix without chemical modification. Two cocrystallizing dyes were employed to validate labeling reproducibility, while a non-cocrystallizing dye served as a negative control to confirm the spatial location of aggregates through complementary contrast. Confocal fluorescence microscopy revealed a stratified distribution of additive aggregates. Numerous small poly(*n*-alkyl acrylates) aggregates (approximately $5\ \mu\text{m}^3$) were densely distributed near the film surfaces. In contrast, larger aggregates (up to $\sim 100\ \mu\text{m}^3$) were fewer in number and with a distribution tending toward the midplane of the film. Upon heating, the fluorescence intensity of the polymer aggregates diminished, corresponding to crystalline melting (onset melting temperature is $40\ ^\circ\text{C}$ according to differential scanning calorimetry) that correlates with adhesive performance changes. Heating the adhesive film transforms the additive aggregates from rigid, load-bearing reinforcements into soft, non-load-bearing inclusions, weakening the cohesive zone at the peel front and thus reducing the peeling force. This work represents the first use of confocal microscopy to directly visualize thermoresponsive additive behavior within a PSA film and provides spatially resolved insights into the mechanism of temperature-triggered adhesive debonding.

KEYWORDS: pressure-sensitive adhesive, confocal microscopy, poly(*n*-alkyl acrylates), cocrystallization, debonding mechanism



INTRODUCTION

Polymers with long alkyl side chains have attracted considerable scientific and technological interest due to their unique architecture, self-assembly, tunable thermal and morphological properties.^{1–3} These materials have been explored in a wide range of applications, including nano-templates, photonic crystals, biological delivery systems, and phase-change materials.^{2,4–9} A key advantage of such polymers lies in their ability to undergo thermally induced phase transitions, which can be precisely controlled by adjusting the length and density of the alkyl side chains. This molecular tunability makes them particularly suitable for temperature-responsive applications, such as shape-memory materials and thermally switchable materials.^{9,10}

Most approaches to developing temperature-responsive adhesives rely on synthesizing new adhesive polymers with tailored thermal properties or designing complex multilayer architectures.^{11–13} While effective, such strategies require reformulation of the base adhesive chemistry and are often

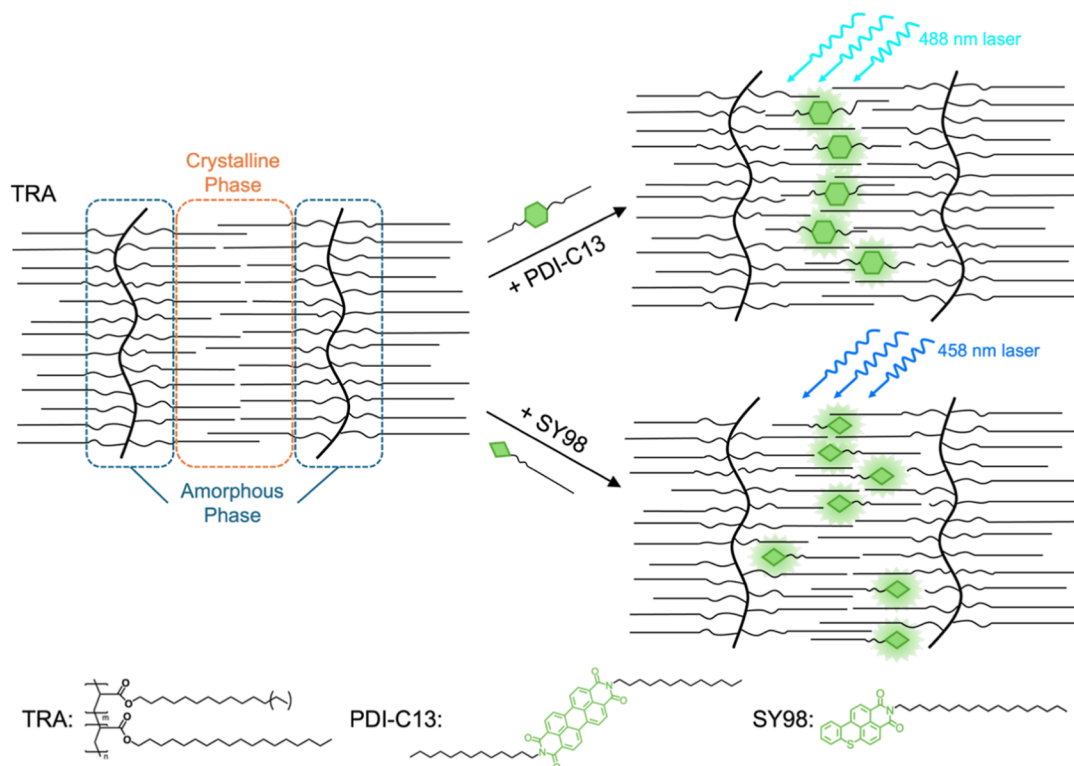
incompatible with existing manufacturing infrastructure. An alternative and more scalable approach involves incorporating temperature-responsive additives into commercial pressure-sensitive adhesive (PSA) formulations. This “drop-in additive” strategy offers several practical advantages: (i) compatibility with existing PSA chemistries without covalent modification, (ii) low additive loading levels (as little as 1 wt %) to minimize cost and potential impairment to adhesive performance, and (iii) seamless adaptability to industrial roll-to-roll (R2R) coating processes. Such an approach is particularly attractive for medical adhesive applications, where material biocompatibility and manufacturing scalability are critical considerations.

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Scheme 1. Schematic Illustration of Noncovalent Fluorescent Labeling via Cocrystallization.



TRA is a semicrystalline polymer whose crystalline phase is formed by ordered packing of its long alkyl side chains, while the main polymer backbone remains amorphous. When dyes bearing long alkyl chains, such as PDI-C13 or SY98, are introduced into the TRA, the alkyl chains of the dyes cocrystallize with the side chains of TRA. As a result, the labeled TRA domains can be selectively excited by the appropriate laser wavelength and visualized using confocal fluorescence microscopy.

In our previous study,¹⁴ we demonstrated that incorporating temperature responsive additives (TRAs)—specifically, poly(*n*-alkyl acrylate) copolymers—into a hybrid rubber-acrylic based PSAs enables a temperature-triggered 75% reduction in peel force. We further demonstrated the scalability of this approach through successful R2R coating trials, confirming compatibility with industrial manufacturing processes.¹⁵ This function is particularly promising for medical applications such as bandages or wound dressings designed for fragile or sensitive skin, and has shown substantial clinical significance in reducing pain and redness during tape removal.^{16,17} However, the underlying mechanism responsible for this thermally activated release remains unclear.

A critical first step toward understanding this mechanism is to investigate the spatial distribution of these polymers with long alkyl side chains within the PSA matrix. In our previous work,¹⁴ we employed surface-sensitive techniques such as atomic force microscopy (AFM) to characterize the adhesive film, revealing nanoscale phase separation at the surface of the thermoresponsive adhesive. However, AFM is inherently limited to surface morphology and provides no information about the polymer distribution through the film thickness.

Several analytical techniques have been developed to characterize additive distribution in PSA films, each with specific capabilities and limitations. Direct analysis in real time time-of-flight mass spectrometry (DART-ToF-MS) has been used to quantitatively analyze additives in pressure-sensitive adhesives,¹⁸ offering high sensitivity for compounds that generate characteristic ionization signatures. However, neutral polymeric additives, such as poly(*n*-alkyl acrylates), lack

distinctive fragmentation patterns or easily ionizable groups, making them difficult to detect and quantify by DART-ToF-MS. Additionally, mass spectrometry techniques provide chemical composition information but do not offer spatially resolved imaging of additive distribution within the film.

Small-angle X-ray scattering (SAXS) is a powerful tool for probing microphase-separated structure in polymer systems and has been applied to study stratification in colloidal and polymer films.^{19,20} Microbeam SAXS with synchrotron sources can achieve depth profiling with micrometer-scale resolution in thin film.²⁰ However, applying SAXS to characterize the distribution of semicrystalline poly(*n*-alkyl acrylate) additives within hybrid rubber-acrylic PSA matrices presents challenges. Both the additive and matrix are primarily composed of low atomic mass elements (C, H, O), resulting in minimal electron density contrast between two phases. While contrast can be enhanced through sample lamination,¹⁹ the chemical similarity between additive and matrix complicates quantitative analysis. Furthermore, SAXS provides ensembled-averaged structural information rather than direct visualization of individual additive domains, and access to synchrotron facilities is required for high-resolution depth profiling.

Given the limitations of existing techniques, laser scanning confocal fluorescence microscopy emerges as a particularly attractive approach for this system. Confocal microscopy is a well-established technique for three-dimensional imaging in biological and materials system^{21–23} and is widely available as a standard laboratory tool.^{24,25} The technique provides sub-micron lateral resolution and axial resolution, enabling nondestructive, high-resolution visualization of internal micro-

structures without physical sectioning.^{21,22} Rapid optical sectioning and z-stack reconstruction allowing precise 3D mapping of polymer distribution throughout the film thickness. When combined with a temperature-controlled stage, confocal microscopy can monitor temperature-induced morphological changes,^{26,27} making it ideal for studying thermoresponsive materials.

Confocal fluorescence microscopy is most effective for optically transparent samples with thicknesses typically under 200 μm , where sufficient laser penetration and resolution can be achieved without significant scattering or absorption.²¹ Our adhesive films are optically clear and typically about 50 μm thick, making them highly compatible with confocal imaging. For applications requiring greater penetration depth or higher resolution, such as imaging thicker (>500 μm) biological tissues or resolving nanoscale features below the diffraction limit, alternative techniques such as two-photon excitation microscopy,²⁸ photoacoustic imaging,²⁹ or sectioned cryogenic electron microscopy may be preferable. However, two-photon microscopy requires specialized instrumentation, specific fluorophores with suitable two-photon absorption and emission characteristics, and is not necessary for transparent, thin-film PSA systems studied here. Similarly, photoacoustic imaging offers penetration depths of several millimeters in highly scattering media but relies on optical absorption contrast rather than fluorescence and, the submicron spatial resolution can be only achieved using specialized research system. Electron microscopy is also challenging as thin sections of polymers films can easily be damaged and distorted by the electron beam. Given the optical clarity, modest thickness and micron-scale aggregate sizes in our PSA films, conventional confocal fluorescence microscopy provides the optimal balance of resolution, accessibility, and functionality for this study.

However, high contrast confocal imaging requires fluorescent labeling to generate optical contrast.²¹ For chemically similar components, such as an acrylic-based additive in a hybrid rubber-acrylic PSA matrix, conventional spectroscopic techniques (e.g., infrared spectroscopy) cannot distinguish the two phases, and fluorescent labeling becomes essential for visualization.

Common labeling strategies include covalent conjugation of dyes to the polymer backbone,^{30,31} nonspecific physical blending of fluorescent dyes,^{32,33} or post-treatment diffusion of dyes into the polymer matrix.³⁴ However, these approaches can be technically demanding, time-consuming, and may interfere with the native properties of the polymer, particularly its microstructure and crystallization behavior. In particular, covalent attachment of dyes, though widely used, often requires multistep chemical synthesis and may alter the thermal or morphological characteristics of the semicrystalline polymer.³⁵ Physical blending or postimpregnation methods may suffer from low labeling specificity or dye migration, resulting in poor spatial resolution and contrast.³²

There is a need for a labeling method that selectively targets the semicrystalline additive domains, preserves its native thermal and morphological properties, and provides sufficient contrast for high-resolution imaging without requiring covalent modification.

In this work, we present a noncovalent, *in situ* fluorescent labeling method that takes advantage of the cocrystallization of long-alkyl-chain fluorescent dyes with the crystalline alkyl side-chain domains of thermoresponsive polymers (Scheme 1). The dyes are simply mixed into the solvent-borne PSA formulation

and selectively cocrystallize with the additive as the solvent evaporates. This approach leverages the structural compatibility between the dye's long alkyl chains (e.g., C13 for PDI-C13) and alkyl side chains of the semicrystalline polymer (C18 and C14), which cocrystallize during film formation. Because the amorphous rubber-acrylic PSA matrix consists of atactic, branched alkyl groups (e.g., 2-ethylhexyl acrylate) that lack the chain length and tacticity required for cocrystallization, the dyes selectively partition into the additive crystalline phase, generating high contrast without covalent attachment or chemical modification.

This labeling method offers several key advantages: (i) it is operationally simple, requiring only mixing of commercially available dyes with the additive prior to coating, and labeling occurs spontaneously during solvent evaporation; (ii) it largely preserves the characteristic crystallization behavior and thermal properties of the additive, with any perturbations from the dye being minor relative to the observable phase transitions; (iii) it provides selective labeling based on physical cocrystallization rather than nonspecific dye diffusion; and (iv) the fluorescence intensity is sensitive to the crystalline-to-amorphous phase transition, enabling monitoring of additive melting behavior. We validated the selectivity and reproducibility of this approach using two chemically distinct cocrystallizing dyes and a noncocrystallizing control.

Using this noncovalent labeling strategy combined with confocal fluorescence microscopy, we investigated the spatial distribution of poly(*n*-alkyl acrylate) additives within PSA films and their temperature-dependent behavior. Our results reveal a stratified distribution of additive aggregates, with small, densely packed aggregates near the film surface and larger, sparser aggregates in the midplane. Upon heating, the aggregates undergo a crystalline-to-amorphous transition that correlates with a reduction in adhesive peel force. These findings provide the first direct microscopic evidence for the mechanism of temperature-triggered debonding: melting of the semicrystalline additive aggregates transforms them from rigid, load-bearing reinforcements into soft, nonload-bearing inclusions, weakening the cohesive zone at the peel front of the adhesive and thus reducing peel strength. This work demonstrates the utility of noncovalent fluorescence labeling for studying thermoresponsive polymer additives, and providing insights into the temperature-triggered debonding mechanism.

MATERIALS AND METHODS

Materials

Two different poly(*n*-alkyl acrylate) copolymers were synthesized as TRAs using procedures similar to those described in our previous work,^{14,15} with minor modifications. One copolymer was synthesized from tetradecyl acrylate (C14) and octadecyl acrylate (C18) in a molar ratio of 27:73. The other was custom synthesized by Scientific Polymers Products Inc. (Ontario, NY) using a molar composition of 81% C18 and 19% dodecyl acrylate (C12) following the same general procedure. The molecular weight parameters of both copolymers are summarized in Table S1.

Throughout this study, the C14/C18 copolymer was used as the TRA for all experiments unless otherwise specified. The only exception was the peel force test, where the C12/C18 copolymer was used due to the unavailability of C14 at the time of sample preparation.

N,N'-Ditridecylperylene-3,4,9,10-tetracarboxylic diimide (PDI-C13, CAS: 95689-92-2) and Nile Red (CAS: 7385-67-3) were purchased from Sigma-Aldrich. Solvent Yellow 98 (SY98) was obtained as a complementary sample from Heubach Group (Heubach

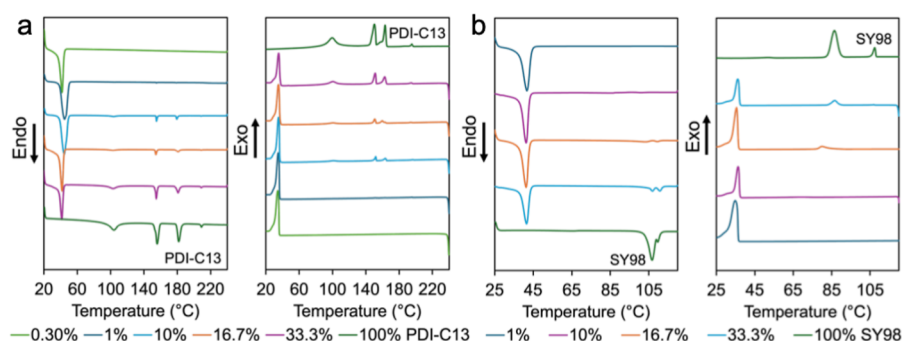


Figure 1. DSC second heating and cooling curve for (a) mixture of TRA and PDI-C13, (b) mixture of TRA and SY98 at different concentrations.

Colorants USA LLC). SY98 is a benzothioxanthene dye with a long 18-carbon alkyl chain.³⁶

The tackified solvent-borne pressure-sensitive adhesive (LOCTITE AH115) was provided by Henkel Corp. It contains a hybrid synthetic rubber-acrylic polymer with a molecular weight specification shown in Table S2. The solvent components and weight fractions are detailed in our previous publication.¹⁵ Nonvolatile (solids) polymers and tackifiers comprise 43.5% of the adhesive.

Sample Preparation

Samples for differential scanning calorimetry (DSC) were prepared by dissolving the desired amount of dye and TRA in toluene, followed by solvent removal in a vacuum oven overnight.

Adhesive films for confocal microscopy were prepared as previously described,¹⁵ with slight modifications. Specifically, dye and TRA were directly dissolved in the solvent-borne PSA at defined concentrations (0.3 wt % dye relative to TRA, 1 wt % TRA in PSA solids). Therefore, the net concentration of dye in the PSA is 0.003 wt %. The solution was cast onto a UV/Ozone treated 2-mil polyethylene terephthalate (PET) film using a sheet-based slot die coater (FOM Technologies, Denmark). The wet films were dried in a static oven at 120 °C for 10 min. Control films were prepared using the same method but without TRA. In the following text, air surface refers to the film surface opposite to the PET backing.

Characterization Techniques

DSC data were collected using a Mettler Toledo DSC 3+ system under a N₂ atmosphere with a heating or cooling rate of 10 °C/min.

Laser scanning confocal microscopy imaging was performed using a ZEISS LSM 710 confocal microscope system and operated using Zeiss Zen 2009 software. All images were acquired using a 63× oil immersion objective with a numerical aperture of 1.4 which provided lateral and axial resolution of approximately 250 and 550 nm, respectively. For three-dimensional imaging, z-stack images were acquired with a slice spacing of 0.9 μm. The published fluorescence excitation and emission spectra for PDI-C13³⁷ and for a 16-carbon alkyl chain benzothioxanthene dye,³⁸ analogous to SY98, were used to select the appropriate confocal microscope excitation laser lines and isolation filters for fluorescence detection. For PDI-C13 stained samples, an excitation laser wavelength of 488 nm was used, and the detector range was set from 520 to 600 nm. For SY98 stained samples, excitation laser wavelength was set as 458 nm and the detector range was set from 500 to 600 nm. For Nile Red-stained samples, an excitation laser wavelength of 561 nm was used, and the detector range was set from 600 to 700 nm. A VAHEAT temperature control system (Interference GmbH) was used to precisely regulate sample temperature and heating rate during imaging.²⁷ Adhesive films were affixed with light pressure to the system's smart substrate (Figure S1) that consists of a custom glass slide with an integrated temperature-control circuit. It is fully compatible with our confocal microscope. This slide also features an integrated thin-film thermocouple positioned near the 5 × 5 mm central sample region, allowing accurate temperature monitoring, and a perimeter-mounted heating element to ensure uniform heating.

Peel force testing was performed using the same homemade peel tester described in our previous publication.^{14,39} In this study, the adhesive samples were prepared by a single transfer coating process using 4.5 μm PET film as the substrate, instead of thicker PET or more deformable films, such as polyurethane films. The use of an ultrathin PET backing minimizes substrate-induced stiffening or elastic recovery effects that could otherwise influence the measured peel force. This approach ensures that the recorded peeling behavior more accurately reflects the adhesive layer's intrinsic properties, rather than artifacts introduced by the backing material.

RESULTS

Cocrystallization Mechanism and Validation

To investigate the cocrystallization behavior between long-alkyl-chain dyes and semicrystalline TRA, a series of dye-TRA mixtures at varying ratios were analyzed using DSC. Figure 1a shows the second heating and cooling thermograms for mixtures of PDI-C13 and TRA, while Figure 1b presents the thermograms for mixtures of SY98 and TRA.

In the case of PDI-C13, when its concentration in the TRA is below 1 wt %—for example, at 0.3 wt %—only a single melting peak and a single crystallization peak are observed. This indicates the presence of a single crystalline phase, suggesting that PDI-C13 has fully cocrystallized with the side chains of the copolymer. As the concentration of PDI-C13 increases, additional melting peaks emerge around 100 °C, 150 °C, 180 °C, and 210 °C, accompanied by multiple crystallization peaks. Comparison with the DSC curve of pure PDI-C13 confirms that these additional transitions correspond to dye-rich crystalline domains formed during the first cycle of heating and cooling, which subsequently melt during the second cycle of heating. These results indicate that at low concentrations, PDI-C13 is molecularly dispersed and cocrystallizes with the polymer side chains, whereas at higher concentrations, excess dye molecules phase-separate and form ordered π - π stacked aggregates, consistent with prior reports on perylene diimide derivatives.³⁷

Similarly, the thermograms in Figure 1b reveal that increasing concentrations of SY98 in the TRA lead to the emergence of multiple melting peaks. However, unlike PDI-C13, SY98 demonstrates better compatibility with TRA. A distinct melting peak associated with SY98 crystallization does not appear until its concentration exceeds 16.7 wt %, significantly higher than the crystallization threshold observed for PDI-C13 (1 wt %). This enhanced miscibility may be attributed to the longer alkyl chains in SY98, which promote more favorable interactions and cocrystallization with the long side chains of TRA.

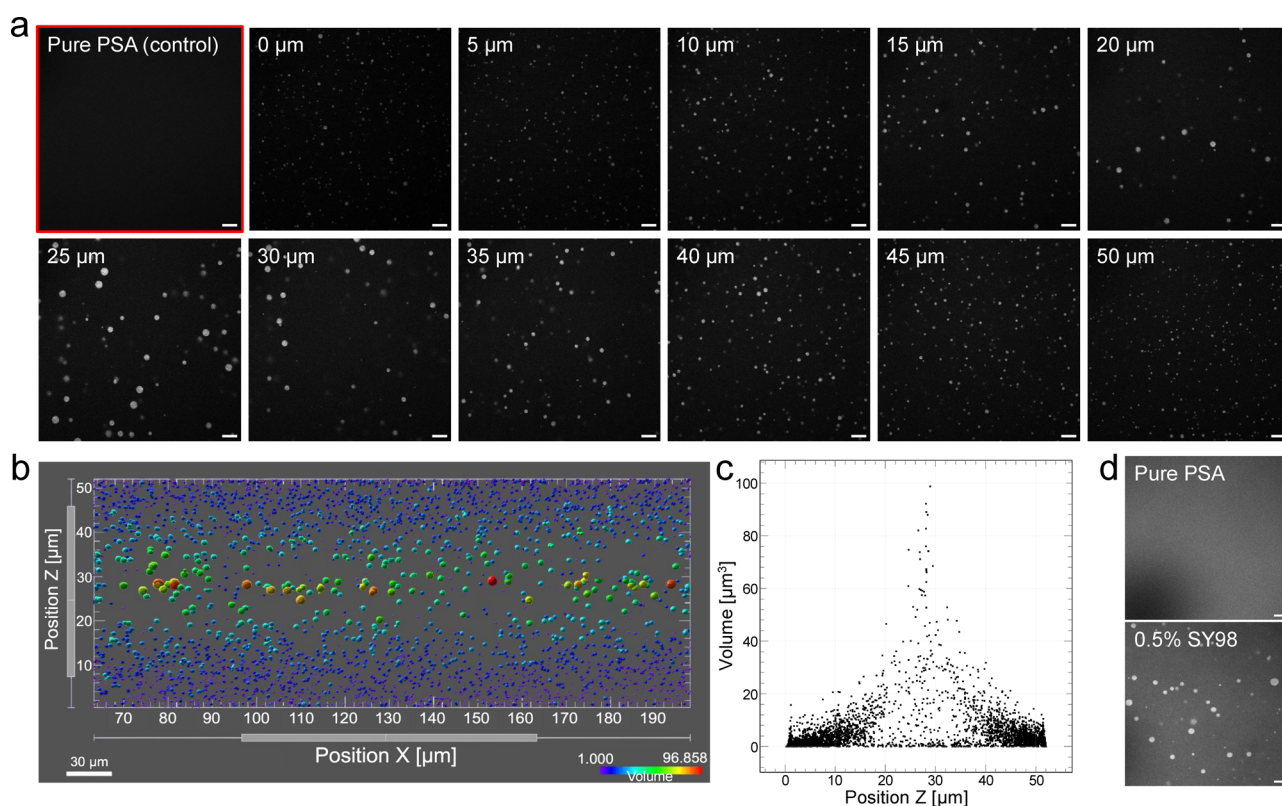


Figure 2. Visualization of TRA distribution within the PSA matrix. (a) Confocal microscopy images of PDI-C13-stained thermoresponsive PSA film (1 wt % TRA) at various depths. The first panel (red border) shows the control sample (pure PSA with 0.003 wt % PDI-C13) at depth of 25 μm . Scale bar: 10 μm . (b) 3D reconstruction of TRA aggregate distribution and volume using confocal z-stack data. The color bar indicates particle volume. (c) Scatter plot showing the volume distribution of TRA aggregates as a function of depth (Z-position) across the PSA film. (d) Confocal microscopy images comparing pure PSA and thermoresponsive PSA (1 wt % TRA) both stained with 0.005 wt % SY98 at depth of 25 μm .

Furthermore, as a perylene diimide derivative, PDI-C13 has a strong propensity for self-aggregation, which hinders its ability to remain amorphous or interact with other molecular species. This self-aggregation likely limits its cocrystallization behavior with the copolymer to low concentrations. On the other hand, PDI-C13 aggregates have very low levels of fluorescence. Therefore, aggregates of dye that are outside the TRA do not produce measurable background fluorescence in the confocal images.

Visualization of TRA Distribution within PSA

To investigate the spatial distribution of TRA within the PSA matrix, we employed confocal fluorescence microscopy. In Figure 2a, the first image (red-framed) shows a control PSA sample without TRA, stained with 0.003 wt % PDI-C13, in which no fluorescent particles were detected, confirming that the fluorescence signal arises specifically from TRA aggregates. This representative image was taken at a depth of approximately 25 μm , and no fluorescence signal was observed throughout the control sample.

A z-stack series of images, as shown in Figure 2a, was captured from a cross-section of the stained PSA film containing 1 wt % TRA and 0.003 wt % PDI-C13, enabling visualization across the film thickness. At the air surface (0 μm), the film exhibits a high density of small aggregates, most of which are below 2 μm in diameter. As the focal plane moves deeper into the film (from 5 to 25 μm), the number density of aggregates decreases, but their average size increases significantly. This trend continues until approximately 25 μm , corresponding to the film's midplane, where the

aggregates reach their maximum size, often exceeding 5 μm in diameter. Beyond the midpoint (30 to 50 μm), closer to the PET substrate, the aggregates gradually become smaller again, the density recovers to air surface level. This distribution indicates a nonuniform aggregation pattern across the film thickness, with large but sparse aggregates in the middle and smaller, densely packed aggregates near the surfaces.

Importantly, this distribution pattern was preserved even when the overall TRA concentration was varied, as demonstrated in Figure S2. This suggests that the thermodynamic or kinetic factors governing TRA aggregate formation during film drying are relatively insensitive to TRA concentration. This suggests that increased film viscosity during film drying may be the controlling factor.

To further quantify the distribution of aggregates, a 3D reconstruction was generated using confocal z-stack data (Figure 2b). Each fluorescent particle was rendered and color-coded according to its volume. The 3D visualization revealed a stratified morphology, with numerous small and densely packed aggregates near the top and bottom surfaces, while larger but sparser aggregates were primarily located near the midplane of the film.

This trend is further supported by quantitative analysis of particle volume as a function of film depth. A scatter plot of aggregate volume versus Z-position (Figure 2c) shows that the largest aggregates form near the midplane of the film (~ 25 μm), while smaller but more densely distributed aggregates are predominantly located near the top and bottom surfaces. This pattern may reflect the balance between solvent evaporation,

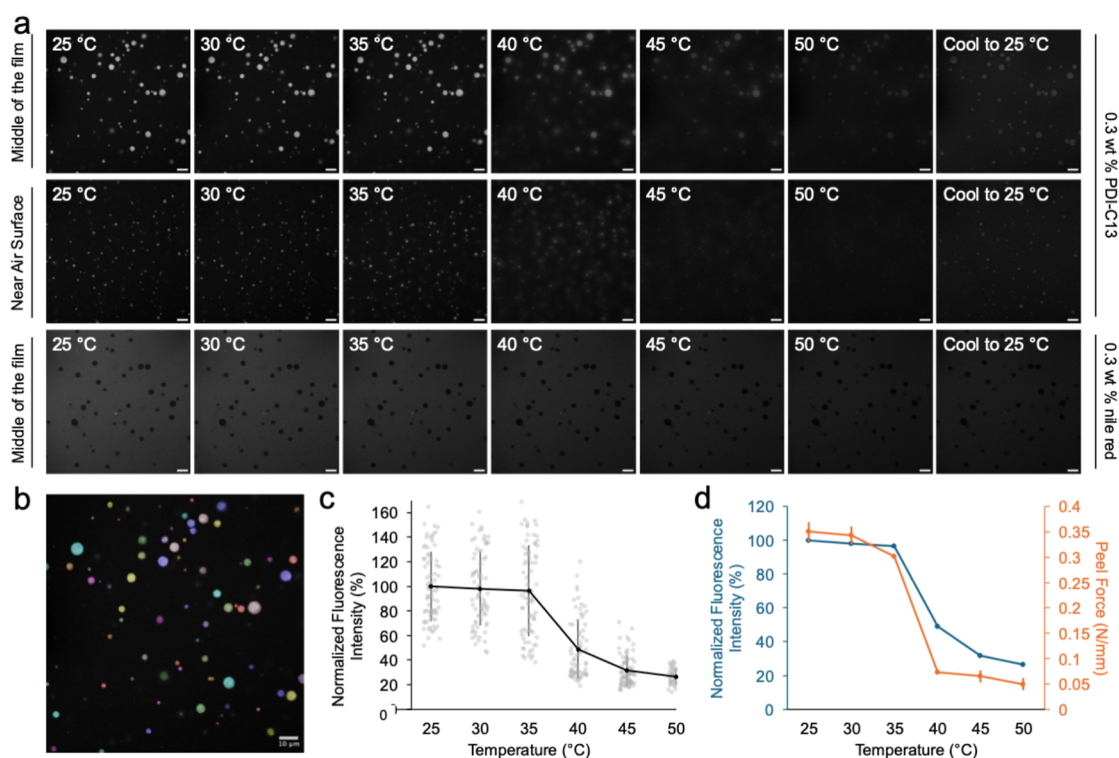


Figure 3. Confocal microscopy images showing changes in TRA aggregates morphology and fluorescence intensity with increasing temperature. (a) Confocal fluorescence images of TRA aggregates within thermoresponsive PSA films stained with PDI-C13 (top two rows) and Nile Red (bottom row), captured at varying temperatures from 25 to 50 °C and after cooling back to 25 °C. The first row corresponds to the middle of the film ($\sim 25 \mu\text{m}$), while the second row captures the region near the air surface ($\sim 5 \mu\text{m}$). Dye concentrations are reported relative to the TRA content; since all films contain 1 wt % TRA, the actual dye concentration in the dry adhesive film is 0.003 wt %. Scale bar: 10 μm . (b) Color coded masks were applied to identify individual TRA aggregates in the confocal images from the first row of Figure 3a. Each color corresponds to a single aggregate, allowing it to be tracked across temperature changes to quantify fluorescence intensity. (c) Normalized fluorescence intensity of TRA aggregates (mean $\pm$ SD, $n = 75$) as a function of temperature, derived from masked particle regions shown in b. (d) Comparison of normalized fluorescence intensity from (c) (blue, left y-axis) and bulk adhesive peel force (orange, right y-axis) as functions of temperature.

increased film viscosity, reduced aggregate diffusion, and crystallization kinetics during film drying.

To verify that this pattern is not specific to a single dye system, a separate set of PSA films was stained with a chemically distinct long-alkyl-chain dye, SY98. As shown in Figure 2d, distinct TRA aggregates were clearly visualized in the SY98-stained thermoresponsive PSA film, while the pure PSA control exhibited only background fluorescence without any visible particulate structures. Both images were captured near the mid plane of the film (approximately 25 μm). Compared to PDI-C13, the SY98-stained samples exhibited lower image contrast. This lower contrast likely arises from the higher miscibility of SY98 with the PSA matrix. Despite the lower contrast, z-stack images of the SY98-stained film, acquired at 5 μm intervals (Figure S3), reveal the same spatial distribution pattern as observed with PDI-C13: smaller, more numerous aggregates near the film surfaces and larger, sparser aggregates in the midplane. This consistency confirms the robustness and general applicability of the cocrystallization-based labeling strategy for tracking TRA distribution in adhesive films.

Changes in TRA Aggregates upon Heating

To further elucidate the thermal behavior of TRA aggregates within the PSA matrix, we conducted confocal fluorescence imaging across a series of temperatures from 25 to 50 °C. Films were stained with either 0.3 wt % PDI-C13 or Nile Red, allowing complementary visualization strategies. As shown in

Figure 3a, both the midplane and near-surface regions were imaged.

In PDI-C13-stained samples (Figure 3a top two rows), TRA aggregates appeared bright and well-defined at room temperature (25 °C) but gradually dimmed and blurred as the temperature increased. Notably, at 40 °C, a distinct fluorescent halo emerged around many aggregates, consistent with the onset of TRA melting. This halo formation suggests that PDI-C13 molecules are partially released from molten TRA aggregates into the surrounding PSA matrix as the dye is no longer retained within the crystalline phase. As the temperature rose to 50 °C, most aggregates lost their bright cores and became indistinguishable from the background, indicating complete melting and significant dye diffusion. After cooling back to 25 °C, the background fluorescence increased by approximately 50% relative to the initial 25 °C image, whereas the aggregate fluorescence recovered by only 11% compared to that at 50 °C. This asymmetric recovery supports partial and irreversible redistribution of dye molecules into the PSA matrix during thermal cycling, which limits fluorescence recovery within the TRA aggregates upon cooling. Photobleaching can be ruled out as a major factor, as PDI-C13 is known for its high photostability. This is supported by Figure S4, which shows that the fluorescence intensity of the PDI-C13 film remains essentially unchanged after 30 min of continuous 488 nm laser scanning.

Conversely, in Nile Red-stained samples, TRA aggregates remained visibly dark against a fluorescent background (Figure 3a bottom row and Figure S5), owing to the known immiscibility between Nile Red and long-alkyl-chain crystallites such as TRA. As previously reported,^{40,41} Nile Red fails to cocrystallize with alkyl chain-rich domains, resulting in exclusion of the dye from the aggregates. This provides a “backlit” view of the aggregates, allowing us to visualize their presence via contrast rather than direct staining. Interestingly, the morphology of these aggregates remained largely unchanged across the temperature range (Figure 3a bottom row), suggesting that the aggregate boundaries remain spatially constrained even during melting.

In both staining systems, TRA aggregates appeared to remain in place during heating, even when fluorescence intensity diminished. Upon cooling back to 25 °C, PDI-C13 fluorescence reappeared at the same locations (Figure 3a, rightmost panels), with the fluorescence intensity recovering to approximately 11% of its original value, indicating that TRA recrystallization occurs without significant lateral diffusion. This observation strongly supports the idea that TRA domains melt *in situ* and reform locally, maintaining their original positions.

This spatial stability of TRA during the melt-recrystallization cycle is consistent with our previous observation of reversible peel force behavior across multiple heating–cooling cycles.¹⁵ The localized melting and recrystallization of TRA aggregates is likely to allow reversible thermal cycling performance.

To quantify the thermal response of TRA aggregates, image processing techniques such as background subtraction, contrast adjustment, and thresholding were applied to confocal micrographs to segment individual aggregates. The resulting particle masks, shown in Figure 3b, were used to extract fluorescence intensity values for each aggregate.

The mean TRA aggregate fluorescence intensity remained nearly unchanged up to 35 °C but decreased sharply by approximately 51% at 40 °C and by about 74% at 50 °C compared to the baseline at 25 °C (Figure 3c), aligning with the melting transition of semicrystalline TRA. Upon cooling back to 25 °C, the fluorescence intensity partially recovered by about 11%. Comparison with bulk adhesive performance revealed a parallel drop in peel force over the same temperature range (Figure 3d), further corroborating that TRA melting governs adhesive weakening at the macroscopic scale.

DISCUSSION

Advantages and Practical Implications

Confocal fluorescence microscopy combined with noncovalent fluorescent labeling via cocrystallization provides a robust and accessible method for visualizing TRA distribution and melting behavior within PSA matrix. This approach employs the long alkyl side-chain domains within TRA to selectively absorb fluorescent dyes with long alkyl chains. In the crystalline state, these rigid domains restrict the rotational and vibrational freedom of the fluorophores, suppressing nonradiative decay pathways and enhancing fluorescence emission. Upon heating above the melting temperature (T_m), TRA aggregate fluorescence diminishes significantly by approximately 74% at 50 °C. We attribute this fluorescence loss to a combination of two process. The first is dye diffusion and an increased solubility of dye in the matrix, supported by the halo-like

feature observed at 40 °C (near the onset of T_m) and a roughly 50% increase in background fluorescence intensity, suggesting partial migration of dye molecules into the surrounding PSA matrix. The second is photophysical relaxation. We speculate that when the TRA transitions into an amorphous state above its melting temperature, the absorbed dyes experience less configuration confinement, allowing greater intramolecular motion and more efficient dissipation of excited-state energy, and results in reduced fluorescence emission. Upon cooling, DSC measurements confirm a reversible recrystallization, however, TRA aggregate fluorescence recovers by only 11%. This limited recovery results from kinetically limited dye redistribution: the diffusion kinetics of the dye in the cooled matrix restrict it from returning to the reforming crystallites at the same rate as TRA recrystallization. Consequently, PDI molecules that remain dispersed in the PSA matrix or form small, subresolution aggregates in the amorphous TRA phase contribute minimal fluorescence. This contrasts with the initial film formation from solution, where solvent-enhanced molecular mobility enabled long-range diffusion and assembly into large, highly fluorescent cocrystallized domains. Overall, we hypothesize that these mechanisms explain the bright TRA aggregates fluorescence at 25 °C, the decrease in TRA fluorescence at elevated temperatures (40 to 50 °C), and the partial recovery, approximately 11% of the original fluorescence intensity, upon cooling back to 25 °C, with the TRA aggregates remaining in relatively unchanged location and volume.

This noncovalent fluorescent labeling technique provides several distinct advantages for studying thermoresponsive pressure-sensitive adhesives. Most importantly, it enables direct visualization of individual additive aggregates rather than relying on ensemble-averaged data. This allows detailed characterization of their morphology, size, and three-dimensional spatial distribution within the adhesive matrix. The observed stratification with smaller, more numerous aggregates near film surfaces and larger, more dispersed ones toward the midplane would be difficult to capture using depth-averaged techniques such as SAXS.

Selective, noncovalent dye labeling is shown to enable *in situ* visualization of the thermal response of a semicrystalline polymer aggregate over a wide temperature range. Variations in fluorescence intensity directly reflect changes in local crystallinity, making it possible to correlate microscopic melting behavior with macroscopic adhesive debonding. This contrasts with bulk thermal methods such as DSC, which provide overall melting information but lack spatial resolution or sensitivity to surface versus interior domains.

Beyond its analytical capability, this method is accessible and scalable. It relies on standard laboratory confocal microscopes and straightforward sample preparation, achieved simply by dissolving long-alkyl-chain dyes and the TRA in the solvent-borne PSA before coating, and labeling will occur spontaneously during solvent drying process. Such simplicity enables rapid screening of formulation parameters, making it practical for industrial and research use alike.

The technique is also noninvasive, relying purely on physical cocrystallization rather than chemical modification. The very low dye loading (~ 0.003 wt %) preserves the inherent crystallization and adhesive properties, while consistent results across multiple dyes (PDI-C13 and SY98) and a noncrystallizing control (Nile Red) confirm the selectivity and reproducibility of the approach.

Finally, this strategy demonstrates broad applicability beyond the current adhesive system. It can be extended to other semicrystalline polymers featuring long alkyl side chains, such as poly(*n*-alkyl methacrylates) and alkyl-functionalized polyolefins, or adapted to study crystallization-driven self-assembly, polymer blend phase separation, and other systems governed by alkyl chain crystallinity.

Potential Limitations

While the proposed labeling and imaging strategy offers notable advantages, its effectiveness is inherently limited by the structural characteristics of the polymer to be labeled and the nature of the fluorescent dye used. The labeling approach relies on the presence of the alkyl side-chain crystallinity within the semicrystalline polymer (Scheme 1), which serves as the physical domain for selective dye incorporation.

In addition, the range of compatible dyes is restricted to those with long alkyl chains capable of cocrystallizing with the semicrystalline polymer. While this method was successfully demonstrated with PDI-C13 and SY98, the pool of available dyes with such structural compatibility is limited.

From an imaging perspective, the resolution of confocal fluorescence microscopy is ultimately constrained by the diffraction limit of light. Although sufficient for resolving micron-scale TRA domains, this resolution may be inadequate for visualizing sub-100 nm features without deconvolution techniques or internal morphology within individual aggregates. For nanoscale structural analysis, higher-resolution techniques such as optical super-resolution microscopy or transmission electron microscopy (TEM) are recommended.

Mechanism Insights into Temperature-Triggered Debonding

Understanding the mechanism for TRA facilitated adhesive debonding upon heating is a central aim of this study. We postulate that the reduction in peel force upon heating can be understood through the framework of multiscale energy dissipation in soft adhesives:

$$\Gamma = \Gamma_0 [1 + f_d(\nu, T)]$$

^{12,42,43}, where Γ is the adhesion energy, Γ_0 is the intrinsic interfacial toughness associated with debonding at the interface, f_d is an intensifying factor determined from bulk energy dissipation that is a function of the crack speed ν and temperature T . This concept originates from the classic work of Gent and Schultz,⁴⁴ who first proposed that peel strength consists of two contributions: an equilibrium work of detachment determined by thermodynamics and an enhancement due to imperfect elasticity of the adhesive. The current empirical form of the equation was later developed by Maugis and Barquins,⁴⁵ who related f_d to rate- and temperature-dependent bulk dissipation.

During peeling of adhesive tapes, energy is dissipated both at the interface (Γ_0), where interfacial bonds break, and in the bulk of the adhesive, quantified by the factor f_d , which reflects energy lost through deformation of material in the cohesive zone. This bulk dissipation arises from large-strain stretching, cavitation, and fibrillation of the adhesive.^{42,43} Importantly, these deformation mechanisms occur well beyond the small-strain elastic regime. As a result, the magnitude of bulk energy dissipation is governed primarily by the strain sustained prior to failure within the cohesive zone, rather than by the small-strain elastic compliance of the adhesive.^{46,47}

The interfacial contribution of Γ_0 can be considered constant, as the surface free energy change of the thermoresponsive adhesive during heating is minimal, and the presence of TRA makes a negligible difference in the surface free energy of the adhesive film, with or without heat (Table S3). Moreover, as shown in Table S3, the measured surface energy remains nearly identical (within experimental limits) for PSAs with and without TRA, both before and after heating, indicating that the temperature dependence of peeling force primarily arises from variations in f_d rather than interfacial effects.

At room temperature, the TRA aggregates are in a solid state and act as load-bearing inclusions within the PSA matrix. Although their presence increases the small-strain stiffness of the adhesive, their more significant effect is to enhance stress transmission and stabilize the cohesive zone near the peeling front. This stabilization delays the onset of cavitation and fibrillation failure, allowing the adhesive to sustain larger strains before rupture. The increased strain-to-failure enables more extensive large-strain deformation within the cohesive zone, leading to greater bulk energy dissipation and a correspondingly higher value of f_d .

When TRA melts at elevated temperature (45 °C), these aggregates become amorphous fluidic inclusions that can no longer bear load. This loss of reinforcement weakens stress transmission within the cohesive zone, promoting failure at lower strains despite the softer local elastic response of the adhesive. Consequently, the extent of large-strain deformation prior to failure is reduced, resulting in diminished bulk energy dissipation and a lower dissipation factor f_d . Thus, the decrease in peeling force upon heating may arise from a weakening of the bulk cohesive response rather than from any change in interfacial bonding.

Although the macroscopic failure mode remains adhesive in both the solid and molten TRA states, the extent of bulk deformation and energy dissipation that precedes interfacial debonding differs substantially between these two conditions. These results demonstrate that thermally induced modulation of peel force arises from changes in large-strain cohesive-zone mechanics rather than from variations in small-strain elastic compliance or interfacial properties.

To further visualize the differences in bulk energy dissipation during peeling, representative peel force–displacement curves are shown in Figure S6. At room temperature, the curves display a higher steady-state peel force and a larger area under the curve, indicating greater mechanical work dissipated prior to failure. Upon heating above the TRA melting temperature, both the plateau peel force and the total mechanical work are reduced. We note that differences in the total displacement between samples may arise from variations in how the tape is mounted and loaded in the peel test, which can affect the effective peeling distance. Importantly, these displacement differences do not affect the observed trend that peeling at elevated temperature consistently results in a lower steady-state peel force. These observations are consistent with a weakening of the cohesive zone at elevated temperature, leading to failure at lower strains despite the softer local elastic response of the adhesive. The peel force–displacement behavior therefore supports the conclusion that thermally induced reductions in peel force arise from diminished large-strain bulk dissipation rather than changes in interfacial adhesion.

Furthermore, considering that the TRA aggregates form via solvent evaporation-induced phase separation, they exhibit inherently low solubility within the PSA matrix. Consequently, even after melting, these aggregates are unlikely to diffuse significantly into the surrounding adhesive, in contrast to the mechanism proposed in a 2020 review¹¹ in which crystalline additives melt and subsequently diffuse into the adhesive matrix, acting as plasticizers that dilute or disrupt the polymer network. In those systems, diffusion of the molten additive leads to permanent matrix modification and is typically associated with irreversible, one-time debonding. Instead, the TRA aggregates remain locally fixed and largely immiscible with the PSA matrix even in the molten state. We believe this spatially localized melting and recrystallization enables repeated thermal cycling without permanent damage to the adhesive matrix, thereby allowing reversible bonding and debonding behavior. This limited mobility is further supported by prior findings on side-chain crystallizable polymers, which show that even in the molten state, long alkyl side chains can retain a degree of nanoscale self-assembly that restricts molecular motion and interdiffusion.⁴⁸

Together, these findings point to a mechanism in which the micron- and submicron-sized TRA aggregates within the PSA matrix undergo a transition from an ordered crystalline state to an amorphous state upon heating, acting as localized, reversible softening agents that reduce the rigidity of the viscoelastic PSA matrix and thereby lower the peel force.

Stratification TRA Distribution During Film Drying

The observed stratified distribution of TRA aggregates across the film thickness arises from a drying process that is inherently asymmetric, as solvent evaporation occurs almost exclusively from the air-facing surface while the opposite side remains in contact with the PET substrate during oven drying. At first glance, such boundary conditions might be expected to produce a strongly asymmetric final morphology. However, the experimental results reveal a near-symmetric stratification characterized by smaller aggregates near both interfaces and larger aggregates in the film interior. This apparent discrepancy indicates that the final aggregate distribution reflects a kinetically arrested, non-equilibrium structure rather than an equilibrium morphology that directly mirrors the drying asymmetry.

Near the air-facing surface, rapid solvent evaporation induces a sharp increase in viscosity and a corresponding reduction in diffusivity during the early stages of drying. Although evaporative solvent flux initially drives TRA transport toward the air interface, the rapid rise in viscosity near the surface kinetically arrests redistribution before extensive aggregation can occur. As a result, TRA aggregation near the air side is arrested at an early stage, leading to the formation of relatively small aggregates in this region.

In contrast, the formation of small aggregates near the PET substrate arises from a different mechanism. Because TRA chains preferentially associate with the solvent phase, evaporative solvent flux toward the air-facing surface can carry a substantial fraction of TRA molecules away from the substrate side and toward the film interior. This transport leads to a relative depletion of TRA concentration near the PET interface, limiting the local availability of TRA chains for aggregation. Meanwhile, the interior of the film retains both higher solvent content and a higher concentration of TRA molecules for a longer duration during drying, providing a

favorable environment for continued TRA mobility and aggregate growth prior to kinetic arrest. The combined effects of early kinetic trapping near the air-facing surface and TRA depletion near the substrate side therefore lead to smaller aggregates at both interfaces and larger aggregates in the PSA interior, producing a near-symmetric stratified distribution despite intrinsically asymmetric drying conditions.

Future Study

To gain further understanding of the switching mechanism of the current adhesive, more dynamic and localized mechanical characterizations are needed. A more rigorous proof consists of mapping the mechanical response of PSA with and without TRA using nanoindentation to access localized PSA stiffness and modulus as a function of temperature. This approach has been demonstrated to quantify localized stiffness of soft polyurethane and rigid polystyrene polymers.^{49,50} By dynamically mapping the mechanical response of PSA regions with and without TRA across different temperatures, it would be possible to determine whether molten TRA domains function as localized softening agents, thereby directly linking their thermal transition to the reduction in peel force.

To capture finer details of TRA morphology, TEM or cryo-TEM^{51–53} could be employed in future work. These techniques would allow high-resolution characterization of TRA domains that are otherwise unresolved by optical methods and could potentially reveal structural motifs or interfacial behaviors critical to the adhesive debonding process.

To further improve 3D visualization of TRA distribution within PSAs, advanced super-resolution fluorescence microscopy techniques, such as stimulated emission depletion (STED) or structured illumination microscopy (SIM),^{54,55} could be explored. This is particularly relevant because the nanoscale surface features observed in our prior AFM studies were not resolved using confocal microscopy, likely due to its limited resolution (~200 to 300 nm laterally). Super-resolution imaging, capable of achieving sub-100 nm resolution,⁵⁵ could bridge this gap by capturing surface and near-surface structures with nanoscale resolution. Beyond static imaging, super-resolution fluorescence techniques also enable dynamic studies, such as monitoring the melting of thermoresponsive additives under thermal stimuli. This represents a key advantage over techniques like TEM which are inherently limited to static, typically cryogenically fixed samples.

CONCLUSION

We developed a non-covalent, *in situ* fluorescent labeling method for visualizing poly(*n*-alkyl acrylates) aggregates embedded within a chemically similar PSA matrix. By utilizing the cocrystallization of long-alkyl-chain dyes with the semi-crystalline polymer, this approach enables clear and complete 3D imaging via confocal fluorescence microscopy. This technique allows for detailed spatial characterization of polymer additive distribution throughout optically transparent adhesive films.

To our knowledge, this is the first application of confocal microscopy with dynamic temperature control to study the morphology and thermal response of additives within PSA systems. The temperature-resolved fluorescence imaging revealed that the thermoresponsive aggregates exhibit melting behavior without distinct diffusion and resolidify upon cooling in their original positions. This approach complements bulk

thermal analysis and provides micron-scale insight into additive behavior, contributing to a more detailed mechanistic understanding of temperature-triggered debonding.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsapm.5c04081>.

Additional experimental details; Molecular weight characteristics of poly(*n*-alkyl acrylate) copolymers and LDT AH115 polymer (Tables S1, S2); Photo of Smart substrate of VAHEAT system (Figure S1); TRA spatial distribution at different concentrations (Figure S2); Confocal microscopy images of thermoresponsive PSA films stained with different dyes (Figures S3–S5); Contact angle and surface energy information on PSA films (Table S3); Peel force–displacement curve at different temperatures (Figure S6) (PDF)

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Notes

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